

While “pleasure” and “enjoyment” are often used to characterize one’s efforts in science; failures, frustrations, and disappointments are equally, if not the more, common ingredients of scientific experience. Overcoming difficulties, undoubtedly, contributes to one’s final enjoyment of success.

S. Chandrasekhar.

“If people do not believe that mathematics is simple, it is only because they do not realize how complicated life is.”

John von Neumann

♣ Following is expected from you.

1. Reach the class room on time.
2. If you have to be absent, find out what was done — before you come to the next class. Also let me know why you were absent.
3. Every week there will be home assignment, please go through, think, work out, pen solutions and Finally read what you have written.
4. If you have trouble understanding an exercise, keep reading again and again. You are sure to succeed. If an exercise asks you to do something complicated, see if you can do a simpler thing and build on your success.
5. Regarding discipline and hard work, stick to your high school routine.
6. Regarding Math, breathe an air of freedom, start thinking and questioning.
7. What is a proof? How to communicate your proof to others?
8. Make a habit to consult books in the library. ♠

1. For $x \in R$, put $|x| = x$ if $x > 0$ and $|x| = -x$ if $x \leq 0$. Show

$$|x_1 + x_2 + \cdots + x_{97}| \leq |x_1| + |x_2| + \cdots + |x_{97}|.$$

- (b) If $x_1 \geq x_2 \geq \cdots \geq x_n \geq 0$ and $y_1 \geq y_2 \geq \cdots \geq y_n \geq 0$, show

$$n \sum x_i y_i \geq \left(\sum x_i \right) \left(\sum y_i \right).$$

- (c) If x, y, z are non-negative numbers, show

$$x^2 + y^2 + z^2 \geq xy + yz + zx, \quad (x + y)(y + z)(z + x) \geq 8xyz,$$

$$x^2 y^2 + y^2 z^2 + z^2 x^2 \geq xyz(x + y + z).$$

2. Let S be a non-empty subset of R . Say that S is bounded below if there is an $a \in R$ such that $a \leq x$ holds for every $x \in S$. Any such number

is called a lower bound for the set S . Greatest lower bound (glb), if exists, is a number l such that it is a lower bound for S and $m \leq l$ holds for every lower bound m of S . glb of S is also called infimum of the set S .

Recall S is bounded above if there is an $b \in R$ such that $x \leq b$ holds for every $x \in S$. Any such number is called an upper bound for the set S . Least Upper bound (lub) of S is also called supremum of the set S .

Say that S is bounded if there is an $a \in R$ and an $b \in R$ such that $a \leq x \leq b$ holds for every $x \in S$.

Show that the lub axiom for R , which we have assumed, implies the following:

- (i) Every non-empty subset of R which is bounded below has a glb.
- (ii) Every non-empty subset of R which is bounded has a glb
- (iii) Every non-empty subset of R which is bounded has a lub.

Show conversely the following: If you did not assume the lub axiom for R , but instead assumed any one of the above three statements then you can prove lub axiom as a theorem.

Moral: We did not show any partiality in assuming lub axiom, it is same as glb axiom.

3. It will be good to have a criterion to recognize lub of a set. Let S be a non-empty set bounded above. Show: A number s is lub of S iff the following two conditions hold: (i) If $x \in S$ then $x \leq s$ and (ii) if $\epsilon > 0$ then there is at least one element $x \in S$ such that $x > s - \epsilon$.
Let S be a non-empty subset of R which is bounded below. A number m is glb of S iff following two conditions hold: (i) If $x \in S$ then $m \leq x$ and (ii) if $\epsilon > 0$, there is at least one element $x \in S$ such that $x < m + \epsilon$.
4. If $a < b$ then we define interval $[a, b] = \{x \in R : a \leq x \leq b\}$.
I have an interval $[a, b]$. Can there be two numbers x, y in this interval such that $x - y > b - a$? Suppose I have two intervals $[a, b]$ and $[c, d]$. Suppose every element of $[a, b]$ belongs to $[c, d]$. In other words $[a, b] \subset [c, d]$. Show that $c \leq a \leq b \leq d$.
5. Let S be a non-empty subset of R . Define a new set, $T = \{-x : x \in S\}$. If S is bounded above then show that T is bounded below. Also show that, if s is lub of S , then $-s$ is glb of T .
6. Let A and B be two non-empty sets of **positive** real numbers. Let us make a new set, $C = \{xy : x \in A, y \in B\}$ If s is lub of A and t is lub of B , show that st is lub of C .

Do you think this will be true if A and B are arbitrary (not necessarily positive) non-empty subsets?

7. Let A and B be two non-empty sets of real numbers. Let us make a new set, $C = \{x + y : x \in A, y \in B\}$. If s is lub of A and t is lub of B , show that $s + t$ is lub of C .
8. Let $x \neq 0$. Find conditions on x so that the set $\{x^n : n \in \mathbb{Z}\}$ is bounded. Find conditions on x so that the set $\{x^n : n \in \mathbb{N}\}$ is bounded.
9. Given a real number $x \in [0, 1]$ show that there exists a sequence of integers $\epsilon_1, \epsilon_2, \dots, \epsilon_n, \dots$ each ϵ_i being one of $0, 1, \dots, 9$ such that

$$-\frac{1}{10^n} \leq x - \left[\frac{\epsilon_1}{10} + \frac{\epsilon_2}{10^2} + \dots + \frac{\epsilon_n}{10^n} \right] \leq \frac{1}{10^n} \quad \forall n$$

We write $x = 0.\epsilon_1\epsilon_2\dots$, decimal expansion of x . We also write

$$x = \frac{\epsilon_1}{10} + \frac{\epsilon_2}{10^2} + \dots + \frac{\epsilon_n}{10^n} + \dots$$

Show that if a number x has two expansions, then one expansion must 'end' with all zeros (called terminating expansion) and the other must end with all 9 (called non-terminating expansion). Further, a number can not have more than two expansions.

10. We have described decimal expansions of numbers in the interval $[0, 1]$. To complete the picture, show that every non-negative integer can be expressed as a finite sum $\eta_k(10)^k + \eta_{k-1}(10)^{k-1} + \dots + \eta_1(10) + \eta_0$ where each η_i is one of the numbers $\{0, 1, 2, \dots, 9\}$. Moreover such an expression is unique subject to $\eta_k \neq 0$, in case $k \geq 1$. We consider only such expressions.

Conclude that if $x \in \mathbb{R}$ and $x \geq 0$, then we can express

$$x = \eta_k(10)^k + \eta_{k-1}(10)^{k-1} + \dots + \eta_0 + \epsilon_1\left(\frac{1}{10}\right) + \dots + \epsilon_j\left(\frac{1}{10}\right)^j + \dots,$$

where each of the numbers ϵ and η are among $\{0, 1, \dots, 9\}$ (with $\eta_k \neq 0$ in case $k \geq 1$). If we were to *denote* $1/10$ by z , and η_i by ϵ_{-i} then this expression takes the pleasing form

$$x = \frac{\epsilon_{-k}}{z^k} + \dots + \frac{\epsilon_{-1}}{z} + \epsilon_0 + \epsilon_1 z + \epsilon_2 z^2 + \dots$$

$$\text{symbolically, } x = \epsilon_{-k}\epsilon_{-k+1}\dots\epsilon_0 \cdot \epsilon_1\epsilon_2\dots\epsilon_j\dots$$

With automatic spellcheck, spelling has deteriorated. With smart phone at fingertips multidigit multiplication in head disappeared. I cannot say whether GPS and navigation software have made my mental maps any more or less accurate. These are some of the skills partially outsourced from many minds, thanks to technological advances. And it will not be long before important aspects of our social and intellectual life follow suit. Forfeiting these skills to gadgets does not worry me so much. What does worry me is the illusion of knowledge and understanding that can result from having information so readily and effortlessly accessible. smarter and more efficient information retrieval via machines could foster dumber and less effective information processing in human minds.

Tania Lombrozo

11. For a number $x > 0$ we defined $\sqrt[n]{x} = \sup\{z > 0 : z^n < x\}$.
Show that $\sqrt[n]{x} = \inf\{z > 0 : z^n > x\}$.
12. Show that x is rational iff it has a recurring decimal expansion.
Show that sum and product of two algebraic numbers is algebraic.
13. Draw the number line and plot the following sets.
(i) $\{x : x^2 - 5x + 6 > 0\}$ (ii) $\{x : x^2 + 5x - 6 \leq 0\}$
(iii) $\{x : -5 < x^4 < 16\}$.
14. I am sure you know the following. Prove them.

$$\sum_1^n i = \frac{n^2}{2} + \frac{n}{2}; \quad \sum_1^n i^4 = \frac{n^5}{5} + \frac{n^4}{2} + \frac{n^3}{3} - \frac{n}{30}.$$

$$\sum_1^n i^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}; \quad \sum_1^n i^3 = \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4}.$$

15. Darshan defines: $(x_n) \rightarrow a$ if $(\forall m \in \mathbb{N}) (\exists n_0) (\forall n \geq n_0) |x_n - a| < 9^{-m}$.
Kabir defines: $(x_n) \rightarrow a$ if $(\forall m \in \mathbb{N}) (\exists n_0) (\forall n \geq n_0) |x_n - a| < 1/m$.
Are these related to our definition? What if they replace $<$ by $\leq$?
16. Suppose that a sequence (x_n) converges. show that the following sequence also converges: $x_1, x_1, x_2, x_2, x_3, x_3, x_4, x_4, x_5, x_5, \dots$.

What is the n -th term of this sequence? What if I repeated each term ten times. What if I repeated n -th term n times, like

$x_1, x_2, x_2, x_3, x_3, x_3, x_4, x_4, x_4, x_5, \dots$

What if I added a few terms at the beginning:

$12, 89, 0, 138, 449, 991, x_1, x_2, x_3, \dots$

What if I deleted a few terms at the beginning: $x_{98}, x_{99}, x_{100}, x_{101}, x_{102}, \dots$?

What if I deleted all odd terms, that is, consider the sequence,

$x_2, x_4, x_6, x_8, x_{10}, x_{12}, \dots$

What if I take integers $1 \leq n_1 < n_2 < n_3 < n_4 < \dots$ and define a sequence by $y_k = x_{n_k}$ for $k = 1, 2, 3, 4, \dots$

17. I have a sequence (x_n) . I know that the sequence $y_n = x_{2n}$ ($n \geq 1$) converges. I know that the sequence $z_n = x_{2n-1}$ ($n \geq 1$) converges. That is, the sequence of even terms converges and the sequence of odd terms converges. Do you think the sequence (x_n) converges. Under what conditions on the limits of these two sequences can you conclude that the sequence (x_n) converges? Can you think of a generalization of the above for finitely many sequences.
18. Suppose a sequence (a_n) of real numbers converges to a number a . Let $P(x)$ be a polynomial of one variable. Show that the sequence of numbers $\{P(a_n)\}$ converges to the number $P(a)$.
19. If $\{x_n\}$ converges, show that $\{|x_n|\}$ converges. Is the converse true?
20. For which real numbers x does the sequence $\{x^n\}$ converge. In such a case, what is the limit?
21. Show

$$\frac{(n+47)^{589}}{2^n} \rightarrow 0; \quad \sqrt[n]{n^{43}} \rightarrow 1; \quad \lim_{n \rightarrow \infty} \sqrt[2n+1]{n^2+n} = 1.$$

22. Show $\sqrt{2}, \sqrt{2+\sqrt{2}}, \sqrt{2+\sqrt{2+\sqrt{2}}}, \dots \rightarrow 2$.
23. Later we shall define the sine function rigorously. But starting with your high school knowledge, discuss the following. For which real numbers x does the sequence $\{(\sin x)^n\}$ converge.
24. If $\sum_{i=1}^{59} a_i = 0$, show that $\lim_{n \rightarrow \infty} \sum_{i=1}^{59} a_i \sqrt{n+i} = 0$. (trivial, tricky)

25. If $a_n > 0$ and $\lim(a_{n+1}/a_n) = L > 0$, then show that $\sqrt[n]{a_n} \rightarrow L$. Use this to evaluate limits of $\sqrt[n]{n}$; $\sqrt[n]{n^5 + n^4}$, $\sqrt[n]{n!/n^n}$. Use the last limit to show that $n! = n^n e^{-n} a_n$ where $\sqrt[n]{a_n} \rightarrow 1$. (Much later, you will learn something called Stirling's formula which gives a better understanding of $n!$).
26. Given a real number $x \in [0, 1]$ show that there exists a sequence $\epsilon_1, \epsilon_2, \dots, \epsilon_n, \dots$ each ϵ_i is either zero or one such that

$$x = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\epsilon_i}{2^i}; \quad \text{written} \quad x = \frac{\epsilon_1}{2} + \frac{\epsilon_2}{2^2} + \dots + \frac{\epsilon_n}{2^n} + \dots.$$

This is called binary expansion of x ; usually written as $\cdot\epsilon_1\epsilon_2\dots$. Show that if a number x has two expansions, then one expansion must 'end' with all zeros (called terminating expansion) and the other must end with all ones (called non-terminating expansion). Further, a number can not have more than two expansions.

27. Let us now fix an integer $r \geq 2$. Let me repeat that this integer is fixed. Given a real number $x \in [0, 1]$ show that there exists a sequence of integers $\epsilon_1, \epsilon_2, \dots, \epsilon_n, \dots$ each ϵ_i being one of $0, 1, \dots, r-1$ such that

$$x = \frac{\epsilon_1}{r} + \frac{\epsilon_2}{r^2} + \dots + \frac{\epsilon_n}{r^n} + \dots.$$

This is called expansion of x to the base r , written $\cdot\epsilon_1\epsilon_2\dots$. Show that if a number has two expansions, then one 'ends' with all zero (terminating) and the other ends with all $r-1$ (non-terminating). Further, a number can not have more than two expansions. $r = 2$ binary expansion; $r = 10$ decimal; $r = 3$ ternary, $r = 16$ hexadecimal. You can understand numbers in several ways, all useful in coding.

I am worried about stupid, it is all around us. Congress (Parliament) debates an issue, both sides appear wrong. Representatives cannot seem to make a reasonable argument. In customer service, the person you are talking to is reading from a script, cannot deviate because they do not know what they are talking about. News has become a mouth piece for views that can be parroted by their listeners. Challenging beliefs is not part of the news anymore. I worry we stopped asking 'why'. We say prescription drug works miracles, but we fail to ask what we really know about what else it does. We glorify stupid on TV shows, showing what dumb things people do, so that we can all laugh at them. I worry behind this glorification of stupidity and the refusal to think hard about real issues are big corporations who make a great deal of money on this. I am worried that people can not think, can not reason from evidence, and do not even know what would constitute evidence. People do not know how to ask the right questions.

Roger Schank

28. We were taught solving quadratic equations, in school with the formula $[-b \pm \sqrt{b^2 - 4ac}]/2a$ which is very important. It is such an excellent answer, we stopped thinking further about the problem. Solve: $x > 0$, $x^2 - x - 1 = 0$. Answer: $x = (1 \pm \sqrt{5})/2$. Let us think afresh. First of all x can not be zero. Want $x^2 = x + 1$. That is, $x = 1 + \frac{1}{x}$. This is very interesting equation: does not give value of x , but explains x in terms of x again which can be used for 'self improvement'. Use this value of x on right side,

$$x = 1 + \frac{1}{1 + \frac{1}{x}}; \quad x = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}}$$

It is natural to believe that the solution is

$$x = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{\ddots}}}$$

Non-sense or meaningful? 'We are the Masters' and we give a meaning. Put $x_0 = 1$, put inductively for $n \geq 1$, $x_n = 1 + \frac{1}{x_{n-1}}$. Shall show the sequence (x_n) , so defined, converges to the solution.

If $\frac{3}{2} \leq x_{n-1} \leq 2$ then show that the same holds for x_n . Conclude $\frac{3}{2} \leq x_n \leq 2$ for all $n \geq 3$, for our sequence. Show

$$|x_{n+1} - x_n| \leq \left(\frac{2}{3}\right)^2 |x_n - x_{n-1}|; \quad n \geq 3.$$

Use this to show (x_n) converges (Cauchy). If the limit is a , show $\frac{3}{2} \leq a \leq 2$ and $a = 1 + \frac{1}{a}$. Thus with high school knowledge $a = (1 + \sqrt{5})/2$. in other words

$$\frac{1 + \sqrt{5}}{2} = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{\ddots}}}$$

29. We have seen that a real number has several dresses — binary, decimal and so on — in which it can appear. For example, consider a , the multiplicative inverse of $1 + 1 + 1$, you can write as $1/3$ or $\frac{1}{3}$ or 3^{-1} . When it wears decimal dress it appears to you as $0.3333333\ldots$. When it wears binary dress, it appears as $0.01010101\ldots$. In ternary dress it has two different styles of appearance $0.100000\ldots$ and $0.0222222222\ldots$. We discuss one more colourful dress possessed by numbers — Ramanujan's favourite.

For any non-negative number a , let $[a]$ and $\langle a \rangle$ denote the largest integer not exceeding a and the fractional part of a respectively. Thus, $\langle a \rangle = a - [a]$.

Fix $x \in (0, 1)$, we define a sequence (finite or infinite) of integers $(n_1, n_2, \dots)$, where each $n_i \geq 1$, as follows. Set

$$n_1 = [1/x]; \quad r_1 = \langle 1/x \rangle = \frac{1}{x} - n_1.$$

In general, having defined n_i and r_i for $1 \leq i \leq k$; put

$$n_{k+1} = [1/r_k]; \quad r_{k+1} = \langle 1/r_k \rangle = \frac{1}{r_k} - n_{k+1}.$$

If at some stage $r_k = 0$, stop and say $[n_1, n_2, \dots, n_k]$ is the **continued fraction expansion** of x . otherwise say $[n_1, n_2, \dots]$ is the continued fraction expansion of x . Accordingly, x has a terminating or non-terminating expansion. We write $x = [n_1, n_2, n_3, \dots]$ or

$$x = \frac{1}{n_1 + \frac{1}{n_2 + \frac{1}{n_3 + \frac{1}{\ddots}}}}.$$

What is the meaning of left side above? If each $a_i > 0$, we define $[a_1, \dots, a_k]$ by induction on k as follows. (pause/think)

$$[a_1] = 1/a_1; \quad \text{and for } k > 1, \quad [a_1, \dots, a_k] = [a_1, \dots, a_{k-2}, a_{k-1} + \frac{1}{a_k}].$$

We define $[n_1, n_2, n_3, \dots] = \lim_k [n_1, \dots, n_k]$. Does this limit exist? Yes, and equals x . We **postpone these matters**, but do the following.

Show that $x \in (0, 1)$ is rational iff the expansion is terminating. What is the expansion for $4/5$?, $144/89$? Which number is $[1, 2, 3, 4]$?

If $x > 1$ and not an integer, then its continued fraction expansion is given by $[n_0; n_1, n_2, \dots]$ where $n_0 = \lfloor x \rfloor$ and $[n_1, n_2, \dots]$ is the expansion of $\langle x \rangle$. Observe n_0 is separated from the rest by a semi-colon, no confusion. If $x \geq 1$ is an integer, we simply say $[x;]$ is its continued fraction expansion (funny?, semicolon followed by bracket).

30. (i) If $0 < a < b$, then show $\sqrt[n]{a^n + b^n} \rightarrow b$.
(ii) If $\nu(n)$ is the number of prime factors of n , show $\frac{\nu(n)}{n} \rightarrow 0$.
(iii) Show $\left(\frac{1}{n^2} + \frac{1}{(n+1)^2} + \dots + \frac{1}{(2n-1)^2} \right) \rightarrow 0$.
(iv) Show $\sqrt[2n+1]{n^2 + n} \rightarrow 1$.
(v) Show that the sequence $\{x_n = \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}\}$ converges.
(vi) Let $a > 0, b > 0$. Put $a_1 = \sqrt{ab}$ and $b_1 = (a+b)/2$. In general $a_{k+1} = \sqrt{a_k b_k}$ and $b_{k+1} = (a_k + b_k)/2$. Show $\{a_n\}$ converges; $\{b_n\}$ converges and the two limits are same, called arithmetic-geometric mean of a, b .
31. Let (x_n) and (y_n) be bounded sequences. Show that
(i) $\liminf(x_n + y_n) \geq \liminf x_n + \liminf y_n$
(ii) $\limsup(x_n + y_n) \leq \limsup x_n + \limsup y_n$.
(iii) $\liminf x_n = -\limsup(-x_n)$. $\limsup x_n = -\liminf(-x_n)$.
(iv) $\limsup(29x_n) = 29 \limsup x_n$. $\liminf(29x_n) = 29 \liminf x_n$.
32. Find $\liminf$ and $\limsup$ of the following sequences
(i) $n^{\sin(n\pi/2)}$ (ii) $\sin(n\pi/3)$
(iii) $\sqrt[n]{3}$ if n odd and $\sqrt[n]{55}$ if n even
(iv) $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ Here $0 < a < b$.
(v) $(1 + \frac{1}{n})^{n+1}$ (vi) $(1 + \frac{1}{n+1})^n$.
(vii) $(1 - \frac{1}{n})^n$ (viii) $(1 + \frac{1}{n})^{2n}$

We have bred docile cows that produce enormous amounts of milk but or otherwise far inferior to their wild ancestors. They are less agile, less curious, and less resourceful. We are now creating tame humans that produce enormous amounts of data and function as very efficient chips in a huge data processing mechanism, but these data cows hardly maximize human potential. Indeed we have no idea what our full human potential is because we know so little about human mind. And yet we do not invest much in exploring mind, instead focusing on increasing the speed of our internet connections and efficiency of our big data algorithms. . . . In the absence of modern methods for the direct mind observation, we might try out some of the tools developed by premodern cultures. Several ancient cultures devoted a lot of attention to the study of the mind, and they relied not on collecting second hand accounts but on training people to observe their own minds systematically. The methods they developed are bunched together under the generic term 'meditation'. Y Harari

33. Do you think a convergent sequence (a_n) must have a maximum, that is, an k such that $a_k \geq a_n$ for all n ? Do you think a convergent sequence (a_n) must have a minimum, that is, an k such that $a_k \leq a_n$ for all n ? Show that a convergent sequence (a_n) must have either a maximum or a minimum.
34. Suppose that $a_n > 0$ for each n and $\liminf a_n = 0$. Show that for infinitely many values of n the following happens:
 a_n is strictly smaller than $a_1, a_2, a_3, \dots, a_{n-1}$.
 Suppose that $a_n > 0$ for each n and the sequence (a_n) converges to zero. Show that for infinitely many values of n the following happens:
 a_n is strictly larger than $a_{n+1}, a_{n+2}, a_{n+3}, \dots$.
35. Let $P(x)$ and $Q(x)$ be two polynomials in one variable x . Assume that $Q(n) \neq 0$ for $n \geq 1$. Discuss convergence of the sequence $\{\frac{P(n)}{Q(n)}\}$.
36. (not easy) If a sequence of numbers is getting close to a then successive averages of those numbers also get close to a . That is, Suppose $x_n \rightarrow a$, put $a_n = \frac{1}{n} \sum_{k=1}^n x_k$ and show that $a_n \rightarrow a$. (check first $a = 0$, after some stage all x_n are too small).
 If a sequence $\{x_n\}$ is such that the sequence of successive averages converges, then the sequence $\{x_n\}$ is said to converge in the sense of *Cesaro*.

Thus any convergent sequence is also convergent in the sense of Cesaro. Show that the converse is not true by considering the ± 1 sequence.

37. Let $a > 0$. Choose a number $x_1 > \sqrt{a}$. Define recursively

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right), \quad n \geq 2$$

Show $x_n \downarrow \sqrt{a}$. If $\epsilon_n = x_n - \sqrt{a}$, show $2\epsilon_{n+1} = \frac{\epsilon_n^2}{x_n} \leq \frac{\epsilon_n^2}{\sqrt{a}}$.

38. Fix $a > 1$. Choose a number $x_1 > \sqrt{a}$. Define recursively

$$x_{n+1} = \frac{a + x_n}{1 + x_n} = x_n + \frac{a - x_n^2}{1 + x_n}.$$

Show $\{x_{2n-1}\} \downarrow$. Show $\{x_{2n}\} \uparrow$. Show $\lim x_n = \sqrt{a}$.

39. If $\sum a_n^2$ converges, show that $\sum \frac{a_n}{n}$ converges. Sometimes a more general formulation gives an idea. If $\sum a_n^2$ and $\sum b_n^2$ converge, then show that the series $\sum a_n b_n$ converges absolutely.

If $\sum |a_n|$ converges, show that $\sum a_n^2$ converges. If $\sum a_n$ converges, do you think that $\sum a_n^2$ converges?

40. Test for convergence of $\sum a_n$ where,

$$a_n = \frac{2n}{2n+1} - \frac{2n-1}{2n}, \quad a_n = (-1)^n \frac{n}{n+1}.$$

41. If $\sum a_n$ and $\sum b_n$ are series of strictly positive terms and $\lim \frac{a_n}{b_n} \rightarrow 1$, show that the series $\sum a_n$ converges iff the series $\sum b_n$ converges. What if the limit were 1000 instead of one.

42. If $0 < r < 1$, you already know that $\sum r^n$ converges. Show that $\sum n^{100} r^n$ converges. More generally, if $P(x)$ is any polynomial in one variable x , show that $\sum P(n)r^n$ converges.

43. Show that for $0 < x < 1$, the series

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

converges.

44. Let $0 < a < b < c < 1$. Show that the following series converges.

$$a + b + c + a^2 + b^2 + c^2 + a^3 + b^3 + c^3 + \dots$$

Show that the series $\sum x_n$ also converges, where $x_n = a^n + b^n + c^n$. Note that this series is different from the series above (just ask yourself: what is the first term, what is the second term). Do these two series have the same limit? Justify your answer.

45. One suggestion to show convergence of the series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

is to write it as

$$(1 - \frac{1}{2}) + (\frac{1}{3} - \frac{1}{4}) + (\frac{1}{5} - \frac{1}{6}) + \dots$$

and show that this series converges. This needs justification, because the two series are different (ask yourself: what is the first term etc). Justify by doing the following. Show that the second series converges. So partial sums are Cauchy. Then argue that for the first series also partial sums are Cauchy.

46. Fix a number h . Let us define $(x)_n$ for $n \geq 0$ by

$$(x)_0 = 1, \quad (x)_n = x(x-h)(x-2h) \cdots (x - [n-1]h).$$

Show the following for $n = 1, 2, 3 \dots$

$$(x+y)_n = (x)_n + \binom{n}{1}(x)_{n-1}(y)_1 + \binom{n}{2}(x)_{n-2}(y)_2 + \cdots + (y)_n.$$

How does this read if $h = 0$?

As I read, I stop and ask: what is the author trying to say? What is the author really thinking? What do I think of this? I talk to myself while reading somebody else's writing. But the main thing is to give myself time, to close my eyes, to give myself space, to reflect and allow my thoughts to form on their own in order to shape my ideas. We humans have a wide range of abilities that help us perceive and analyze mathematical content. We perceive abstract notions not just through seeing but also by hearing, by feeling, by our sense of body motion and position. Our geometric and spatial skills are highly trainable. A whole mind approach to mathematical thinking is vastly more effective than the common approach that only manipulates symbols.

William Thurston

We are now going beyond high school. Think, do not give up.
Remember 'Home Alone' – an idea, a plan and then execution.

47. Test for convergence, Use your knowledge about log and sine functions.

$$\begin{aligned} & \sum \frac{2000}{\sqrt[3]{35n^4 - 29}}; & \sum \frac{2000}{\sqrt[4]{35n^3 - 29}}; \\ & \sum \frac{1}{n(1.01)^n}; & \sum \frac{\log n}{n^{1.0001}}; & \sum \frac{n^{100000}}{n!}; & \sum \frac{(100000)^n}{n!}. \\ & \sum_{n \geq 100} \frac{1}{n \log n (\log \log n)^p} & \sum_{n \geq 10000} \frac{1}{n \log n \log \log n (\log \log \log n)^p} \\ & \sum (-1)^n \frac{\sin(19/n)}{\log \log(100 + n)}. & \sum \frac{1}{n} \sin \frac{1}{n}. \\ & \sum_{n \geq 2} \frac{\log(n+1) - \log n}{(\log n)^2}; & \sum_{n \geq 1} \frac{1 \cdot 2 \cdot 3 \cdots n}{(\alpha + 1)(\alpha + 2) \cdots (\alpha + n)}. \quad (\alpha > 0) \end{aligned}$$

48. Sometimes the statement of the problem is long and probably frightening, but the solution is immediate. Here are two such problems.

Let $\sum a_n$ be a series of non-zero numbers. Suppose that there is an $\epsilon > 0$ and an integer n_0 such that $(\clubsuit)$ holds for every $n \geq n_0$ then the series $\sum a_n$ converges absolutely. If $(\spadesuit)$ holds for all $n \geq n_0$ then it does not converge absolutely.

$$\frac{\log \frac{1}{|a_n|}}{\log n} > 1 + \epsilon. (\clubsuit) \qquad \frac{\log \frac{1}{|a_n|}}{\log n} < 1 - \epsilon. (\spadesuit)$$

49. We know If $a_n \downarrow 0$, then $\sum_{n \geq 1} (-1)^{n-1} a_n$ converges. Here is a generalization. Let $\sum b_n$ be a series with bounded partial sums. Let $a_n \downarrow 0$. Then the series $\sum a_n b_n$ converges. Note: Not demanding convergence of $\sum b_n$. By taking $\sum b_n = \sum \pm 1$ series, you get the theorem on alternating series.

use: if (s_n) are partial sums of the series $\sum b_n$, then for $m > n$

$$\begin{aligned} a_{n+1}b_{n+1} + a_{n+2}b_{n+2} + a_{n+3}b_{n+3} + \cdots + a_{m-1}b_{m-1} + a_mb_m = \\ -s_na_{n+1} + s_{n+1}(a_{n+1} - a_{n+2}) + s_{n+2}(a_{n+2} - a_{n+3}) \\ + \cdots + s_{m-1}(a_{m-1} - a_m) + s_ma_m. \end{aligned}$$

Application: show that the series $\sum \sin nx$ has bounded partial sums. Conclude that $\sum \frac{\sin nx}{n}$ converges.

50. We have shown that Cauchy product of two absolutely convergent series is convergent. Show that it is absolutely convergent.
51. If $a_n \geq 0$ for all n and $\sum a_n$ converges, then so does $\sum \frac{\sqrt{a_n}}{n}$.
52. $\sum a_n$ is a series of positive numbers which does not converge. Show that $\sum \frac{a_n}{1+a_n}$ does not converge.

If $s_n = a_1 + \cdots + a_n$, show that

$$\frac{a_{n+1}}{s_{n+1}} + \cdots + \frac{a_{n+k}}{s_{n+k}} \geq 1 - \frac{s_n}{s_{n+k}}$$

Deduce that $\sum \frac{a_n}{s_n}$ does not converge.

Show that

$$\frac{a_n}{s_n^2} \leq \frac{1}{s_{n-1}} - \frac{1}{s_n}.$$

Deduce that $\sum \frac{a_n}{s_n^2}$ converges.

Discuss convergence of

$$\sum \frac{a_n}{1+na_n} \quad \sum \frac{a_n}{1+n^2a_n}.$$

53. $\sum a_n$ is a convergent series of strictly positive numbers. Put $r_n = \sum_{m \geq n} a_m$. If $m < n$ show that

$$\frac{a_m}{r_m} + \cdots + \frac{a_n}{r_n} > 1 - \frac{r_n}{r_m}.$$

Deduce that $\sum \frac{a_n}{r_n}$ does not converge.

Show that

$$\frac{a_n}{\sqrt{r_n}} \leq 2(\sqrt{r_n} - \sqrt{r_{n+1}}).$$

Deduce that $\sum \frac{a_n}{\sqrt{r_n}}$ converges.

54. $a_1, a_2, a_3, \dots, a_{k-1}, a_k$ are distinct nonzero reals. Show

$$\begin{aligned} & \frac{1}{a_1} - \frac{1}{a_2} + \frac{1}{a_3} - \dots \pm \frac{1}{a_k} \\ &= \frac{1}{a_1 + \frac{a_1^2}{a_2 - a_1 + \frac{a_2^2}{a_3 - a_2 + \frac{a_3^2}{\ddots + \frac{a_{k-1}^2}{a_k - a_{k-1}}}}} \end{aligned}$$

Later we show

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \dots$$

Conclude

$$= \frac{1}{1 + \frac{1^2}{1 + \frac{2^2}{1 + \frac{3^2}{1 + \frac{4^2}{1 + \ddots}}}}}$$

Later we show

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots$$

conclude

$$\frac{4}{\pi} = 1 + \frac{1^2}{2 + \frac{3^2}{2 + \frac{5^2}{2 + \frac{7^2}{2 + \ddots}}}}$$

Do you see beauty in these formulae?

(Paul Loya: Amazing and Aesthetic Aspects of Analysis)

The quest for uniformity is the worst enemy of thinking, creativity, understanding, and open knowledge, which can thrive only in open societies. Indeed, diversity and differentiation are an integral part of the quest for excellence in higher education. Students enter the world of higher education to learn. Understanding existing knowledge is the first step. The ability to ask relevant questions, a capacity to critique conventional wisdoms, and the confidence to resist the authority of the spoken - even printed - words are the next successive steps. Learning is a continuous process and never stops.

The unfolding reality in India is disturbing. Curricula are regulated and straightjacketed. Readings are prescribed. Even the freedom for teachers and students to organize discussions or debates on campus which are perceived as critical of the government is circumscribed everywhere, often by administrative fiat or penal action.

The absence of academic freedom in universities will inevitably harm the teaching-learning process just as it will stifle thinking and creativity in research. Students and teachers will obviously be the worse off. Ultimately, however, the economy, society and polity will be the losers

Deepak Nayyar, (14-08-2025, Hindu)

55. Prove by induction:

$$(1+x)(1+x^2)(1+x^4)\cdots(1+x^{2^{n-1}}) = \frac{1-x^{2^n}}{(1-x)} \quad x \neq 1$$

$$1+2x+3x^2+\cdots+nx^{n-1} = \frac{1-(n+1)x^n+nx^{n+1}}{(1-x)^2} \quad x \neq 1$$

56. Consider the sequence:

$$\frac{1}{1}; \frac{3}{2}; \frac{7}{5}; \frac{17}{12}; \cdots \frac{p_n}{q_n}; \cdots$$

where $p_{n+1} = p_n + 2q_n$ and $q_{n+1} = p_n + q_n$. Show that for all n , the fraction is in lowest terms (no common factors numerator/denominator). Show that the sequence converges to $\sqrt{2}$ and error between the term and $\sqrt{2}$ alternates in sign.(?)

57. consider any integer $b \in N$ and a be the integer closest to $b\sqrt{2}$. Let $(a + b\sqrt{2})^n = a_n + b_n\sqrt{2}$. Show that a_n is the integer closest to $b_n\sqrt{2}$. (Think of an idea first for this and next problem.)
58. For $n \in N$, show there is $k \in N$ so that $(\sqrt{2} - 1)^n = \sqrt{k} - \sqrt{k-1}$.
59. Show (by induction): if $n \in N$ and $n > 1$ then n is a product of primes. That is, either n is prime or there are primes whose product is n .
 $n \in N$ is prime if (i) $n > 1$ and (ii) $n = pq$ with $p, q \in N$ implies either $p = 1$ or $q = 1$.
60. Consider the sequence

$$x_1 = \frac{2^2}{1 \times 3}, \quad x_2 = \frac{2^2}{1 \times 3} \frac{3^2}{2 \times 4}, \quad x_3 = \frac{2^2}{1 \times 3} \frac{3^2}{2 \times 4} \frac{4^2}{3 \times 5}, \dots$$

Show $x_n = 2(n+1)/(n+2)$ and find $\lim x_n$.

61. Find radius of convergence:

$$\frac{x}{1 + \sqrt{1}} + \frac{x^2}{2 + \sqrt{2}} + \frac{x^3}{3 + \sqrt{3}} + \dots + \frac{x^n}{n + \sqrt{n}} + \dots$$

$$x + \frac{2^k}{2!}x^2 + \frac{3^k}{3!}x^3 + \frac{4^k}{4!}x^4 + \dots + \frac{n^k}{n!}x^n + \dots$$

$$x + \frac{2!}{2^2}x^2 + \frac{3!}{3^3}x^3 + \frac{4!}{4^4}x^4 + \dots + \frac{n!}{n^n}x^n + \dots$$

$$\sum_0^{\infty} \frac{x^n}{a^n + b^n}; \quad \sum_0^{\infty} \left(\frac{a^n}{n} + \frac{b^n}{n^2} \right) x^n \quad a > 0, b > 0$$

$$\sum_1^{\infty} \left(1 + \frac{1}{n} \right)^{n^2} x^n; \quad \sum_1^{\infty} n^{-\sqrt{n}} x^n; \quad \sum_1^{\infty} \frac{n!^2}{(2n)!} x^n$$

62. If $P(x) = \sum_0^{\infty} a_n x^n$ has radius of convergence R ; find radius of convergence for $Q(x) = \sum_1^{\infty} n a_n x^{n-1}$; $P(x) = \sum_2^{\infty} n(n-1) a_n x^{n-2}$
63. **comparison test:** $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges.
Ratio comparison test: (a_n) and (b_n) positive numbers and $\frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_n}$ for all n . if $\sum b_n$ converges, show $\sum a_n$ converges.
 You know $\sum (1/n)$ does not converge. Let us enumerate, as $\{a_n, n \geq 1\}$, all the integers in whose decimal representation the digit 4 does not appear. show $\sum (1/a_n)$ converges.

64. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = 0$ if x is rational and $f(x) = 1$ if x is irrational. Find the set of all points a such that f is continuous at a .
Do the same thing for the following functions:
 (a) $f(x) = 0$ if x is an integer and $f(x) = 1$ when x is not an integer.
 (b) $f(x) = 1$ if $x \geq 0$ and $f(x) = 55$ if $x < 0$.
 (c) $f(x) = 1$ if $x > 0$; $f(x) = 55$ if $x < 0$ and $f(0) = 44$.
 (d) $f(x) = \lfloor x \rfloor$ (e) $f(x) = x \lfloor x \rfloor$ (f) $f(x) = \lfloor 1/x \rfloor$ on $(0, \infty)$.
 (g) $f(x) = \langle x \rangle$, the fractional part of x .
65. You know that the function $f(x) = x^2$ is continuous. If I give $a = 2$ and $\epsilon = 1$ what will be your δ ? What if $a = 20$ and same ϵ . What if $a = 200$ and same ϵ ?
Let $f(x) = 1/x$ defined on the interval $(0, 33)$. Take $\epsilon = 0.1$ and $a = 5$, find δ . With the same ϵ and $a = 1$, $a = 1/5$ and $a = 1/100$ find δ .
66. Let $f(x)$ be defined on the real line as follows: if x is irrational number then $f(x) = 0$. If x is a rational number p/q in lowest terms, then $f(x) = 1/q$. Show that f is continuous at a iff a is irrational number.
 p/q in lowest terms means common factors of p, q are ± 1 .
Sometimes this function is called Karl Thomae's function or modified Dirichlet function.
67. Using ϵ - δ definition, show that $f(x) = \sqrt{x}$ is continuous on $[0, 1]$. Do the same for $f(x) = 31x^3 + 29x^2 + 40x + 11109$ at $x = 2$.
68. Let $f : [31, 103] \rightarrow [31, 103]$ be a continuous function. Show that there is a number a such that $f(a) = a$.
69. f and g are two continuous functions. I know $f(x) = g(x)$ whenever x is rational. Show $f(x) = g(x)$ for all x . What if I knew the hypothesis only for rational numbers $x = \frac{m}{n}$ in lowest terms with $n \geq 100$.
70. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function and if $f(x+y) = f(x) + f(y)$ for every x, y then show that there is a number c such that $f(x) = cx$ for all x .
71. Show that $f(x) = 1/x$ is a continuous function on $[1, \infty)$. Show that given $\epsilon > 0$, there is a $\delta > 0$ that works for all x . That is whatever x you take, if $|y - x| < \delta$ then $|f(x) - f(y)| < \epsilon$.

Creativity is present in each and everyone of us due to us being humans and having the ability to think and innovate. It is basic human nature to be curious, to try new things, and to learn by discovering new connections. Somehow, along the way, this natural creative talent has been blocked. Through self-judgment and the conditioning of others, people stop looking for new ideas, stop trying new approaches, and stop discovering new connections. Over time, their creative mindsets weaken and in some cases, may even atrophy to the point that, when called upon, the creative mind-set is so weak they are not able to jump into action.

Creative thinking involves leaving the known and entering the unknown. It is about burning new neural connections in the brain so that creative ideas have the opportunity to surface. Many people stop themselves from experiencing new things and only want to stay in their comfort zones.

Aditya Shanbhag

72. Let $a_1, a_2, \dots, a_n$ be strictly positive numbers.
 Show: $\sqrt[n]{a_1 a_2 \cdots a_n} \leq (a_1 + a_2 + \cdots + a_n)/n$.
 When n is an integer of the form $2, 2^2, 2^4, 2^8, 2^{16}, \dots$ (the exponent of 2 is itself a power of 2), show that $\sqrt[n]{a_1 a_2 \cdots a_n} \leq \frac{a_1 + a_2 + \cdots + a_n}{n}$. Actually this is true for all integers $n \geq 2$. You need not do.
 Idea for $n = 3$: Have a_1, a_2, a_3 , take $a_4 = \sqrt[3]{a_1 a_2 a_3}$ and try your luck.
73. If f is a monotone function defined on an interval and has the intermediate value property, show that f is continuous. What does $f(x) = \sin(1/x)$, $f(0) = 0$ teach you?
74. Let $f(x)$ be defined as follows: if x is irrational then $f(x) = 0$. If x is rational $x = p/q$ in lowest terms, then $f(x) = 1/q$. Show that f is continuous at a iff a is irrational number.
75. Let $r_1, r_2, \dots$ be an enumeration of the set of rationals on the real line. For each $x \in \mathbb{R}$, let $f(x) = \sum_{n: r_n \leq x} \frac{1}{2^n}$. Show f is continuous at a point a iff a is irrational.
 Let $D \subset \mathbb{R}$ be a countable set. Find a function on $\mathbb{R}$ whose set of discontinuity points is exactly D .
76. Let ABC be a triangle in the x - y plane (A, B, C not collinear). Given a line L , show there is a line parallel to L which bisects the triangle into two parts of equal area.

77. Using Cauchy product of series, show:

$$\sin(x+y) = \sin x \cos y + \cos x \sin y, \quad \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

78. Show that the functions $f(x) = \sqrt{x^2 - 1}$ defined on $(1, \infty)$ and $g(y) = \sqrt{y^2 + 1}$ defined on $(0, \infty)$ are inverses of each other.

79. Find the radius of convergence of the following power series (ratio test is better than the formula).

$$\sum \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} x^n$$

$$\sum \frac{(2n)!}{n!n!} x^n. \quad \sum \frac{(n+1)! - n!}{2^n} x^n. \quad \sum \frac{(n!)^2}{(n^2)!} x^n. \quad \sum \frac{n!}{n^n} x^n.$$

$$\sum \frac{e^{n^2}}{n!} x^n. \quad \sum \frac{n^3}{e^n} x^n. \quad \sum \frac{e^n}{n!} x^n. \quad \sum \frac{2^n}{(n!)^{\sqrt{2}}} x^n.$$

80. Show: For $x \neq \pm 1$.

$$\frac{x}{1-x^2} + \frac{x^2}{1-x^4} + \frac{x^4}{1-x^8} + \frac{x^8}{1-x^{16}} + \cdots$$

converges to $x/(1-x)$ when $|x| < 1$ and converges to $1/(1-x)$ when $|x| > 1$. (hint: sum upto n terms = $1/(1-x)$ minus $1/(1-x^{2^n})$).

81. Find $\lim_{x \rightarrow 0 \pm} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$ $\lim_{x \rightarrow 0 \pm} \frac{\sin x}{x}$ $\lim_{x \rightarrow 0 \pm} \frac{1 - \cos x}{x^2}$.

Sine and cosine functions are defined by power series.

Find $\lim_{x \rightarrow 0+} x \log x$ $\lim_{x \rightarrow 0+} x^{0.001} \log x$

82. If $a_n \downarrow 0$ and $\sum a_n$ converges, then show $na_n \rightarrow 0$.

(hint: Given $\epsilon > 0$ choose and fix M verifying Cauchy criterion for $\epsilon/2$.

Choose $N > M$ so that $a_n < \epsilon/(2M)$ for $n \geq N$.)

83. $a > 0$ and $b > a + 1$. Show the following series

$$1 + \frac{a}{b} + \frac{a(a+1)}{b(b+1)} + \frac{a(a+1)(a+2)}{b(b+1)(b+2)} + \frac{a(a+1)(a+2)(a+3)}{b(b+1)(b+2)(b+3)} + \cdots$$

converges to $(b-1)/(b-a-1)$. (hint: If the series is denoted $u_0 + u_1 + u_2 + \cdots$ then $(b+n)u_{n+1} = (a+n)u_n$ so that $(b-a-1)u_{n+1} = (a+n)u_n - (a+n+1)u_{n+1}$ to deduce convergence. To find limit argue $nu_n \rightarrow 0$). Slightly more complicated series (Gauss Hypergeometric series) arise in several applications and are difficult to handle.

84. f is a continuous function on $(0, 1]$. Show it is uniformly continuous iff $f(0^+)$ exists (we can talk about right limit at 0).
85. (Weierstrass M -test) There are numbers M_n such that $|f_n| \leq M_n$ on $[a, b]$ and the series of numbers $\sum M_n$ is convergent. Show that $\sum f_n$ is uniformly convergent.
86. Show that a polynomial of odd degree has at least one root (Fundamental theorem of algebra not allowed)
87. A one-one continuous function on $[0, 1]$ is monotone. Show.
88. There are hard to imagine beautiful continuous monotone functions f . Define a sequence of functions f_n on R as follows: Each of the functions is 0 on $(-\infty, 0)$ and is 1 on $(1, \infty)$. Here is the definition on $[0, 1]$:
- f_0 (its graph) is the straight line joining the points $(0, 0)$ and $(1, 1)$.
- f_1 (its graph) is the polygonal line joining the points $(0, 0)$, $(\frac{1}{3}, \frac{1}{2})$, $(\frac{2}{3}, \frac{1}{2})$, $(1, 1)$. Thus f_1 is flat on middle 1/3-rd interval and increases uniformly on two intervals, left and right (LR).
- f_2 (graph) joins $(0, 0)$, $(\frac{1}{9}, \frac{1}{4})$, $(\frac{2}{9}, \frac{1}{4})$, $(\frac{1}{3}, \frac{1}{2})$, $(\frac{2}{3}, \frac{1}{2})$, $(\frac{7}{9}, \frac{3}{4})$, $(\frac{8}{9}, \frac{3}{4})$ and $(1, 1)$. Thus f_2 is flat on the middle 1/3-rds of the preceding two intervals (L,R) uniformly increasing in the four intervals (LL,LR, RL, RR).
- In general If f_n ($n \geq 1$) is obtained Get f_{n+1} as follows. Keep it unchanged in the intervals where f_n was flat; in each of the other 2^n intervals make it flat in their middled one-third and increasing uniformly to its left and right. Do you get the picture?
- Show: the sequence $\{f_n\}$ converges uniformly and hence the limit function f is continuous; graph of f on $[0, 1]$ is called (try to guess why) Devil's stair case.

Most of our problems fall into one of two categories: either we know what needs to be done and we are not doing it or we really just do not know what to do. Asking yourself ‘what is my purpose?’ is a good way to check on whether what you are doing is really what needs to be done. But many of us have such busy lives that we do not take time to reflect on what we are doing, we go from one task to the next without pausing to consider what is truly important. There are many tasks to do that are urgent that we can easily spend all our time going from one urgent matter to the next without devoting any time to things that are very important.

Gregg Krech

89. Suppose $f : R \rightarrow R$ is differentiable and f' is a polynomial of degree $n - 1$. Show that f must be a polynomial of degree n .
90. Let $f(x) = x^2 + bx + c$. Show that f is increasing for $x > -b/2$ and decreasing for $x < -b/2$.
91. Let $f : R \rightarrow R$ be a function. Then f is said to be an even function if $f(-x) = f(x)$ for every $x \in R$. f is said to be an odd function if $f(-x) = -f(x)$.
Give two examples of even functions and two examples of odd functions. If f is differentiable and is even, show that f' is an odd function. If f is differentiable and is odd, then show that f' is an even function. If f' is odd show that f is even function. If f' is even and $f(0) = 0$, show that f is odd.
92. Find the point of intersection of: tangent to the curve $y = x^2 - x$ at the point $(2, 2)$ and tangent to the curve $y = 1 - x^2$ at the point $(1, 0)$.
93. Show that the function $f(x) = x^2 + 3x + 1$ defined on the interval $(1, \infty)$ is invertible. Find its inverse g . Verify $f'(g(y))g'(y) = 1$; $g'(f(x))f'(x) = 1$. for every $x \in (0, 1)$.

94. calculate

$$\lim_{x \rightarrow 0} \frac{1}{x} e^{-1/x^2}.$$

Let $P(x)$ be any polynomial. Show that $P(x)e^{-x} \rightarrow 0$ as $x \rightarrow \infty$.

95. f is defined on the interval $[-1, 1]$ and for numbers $x, y \in [-1, 1]$.
 $|f(x) - f(y)| \leq 589 \sin(|x - y|^{235})$ Show that f is a continuous function.
 Show that f is differentiable. Show that f is a constant function.
96. Let f be a continuous function on the interval $[0, 1]$ which is differentiable at every point of $(0, 1)$. Assume that $|f'(x)| \leq 33$ for every $x \in (0, 1)$. Show that $|f(x) - f(y)| \leq 33|x - y|$; $x, y \in [0, 1]$.
97. Since you are learning linear algebra, here is an interesting formula. suppose I have nine differentiable functions $\{f_{ij}; 1 \leq i, j \leq 3\}$ on R . Define a function φ on R by

$$\varphi(x) = \begin{vmatrix} f_{11}(x) & f_{12}(x) & f_{13}(x) \\ f_{21}(x) & f_{22}(x) & f_{23}(x) \\ f_{31}(x) & f_{32}(x) & f_{33}(x) \end{vmatrix}$$

Here $|A|$ is determinant of A . Show that φ is differentiable and

$$\varphi' = \begin{vmatrix} f'_{11} & f'_{12} & f'_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix} + \begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f'_{21} & f'_{22} & f'_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix} + \begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f'_{31} & f'_{32} & f'_{33} \end{vmatrix}$$

98. Find derivatives of the following functions.

$$e^{(x^{33})}. \quad (e^x)^{33}. \quad \sin(\cos x). \quad \frac{e^{\cos x}}{1 + x^2}.$$

99. Let $f(x) = -x \log x - (1 - x) \log(1 - x)$ ($0 < x < 1$). Show that it is possible to define this function at the points zero and one so that it is continuous on the closed interval $[0, 1]$? Where is the function increasing, where is it decreasing, Where is its maximum value, what is it? Sketch its graph.
100. Prove the following formula, called Leibnitz's formula. $f^{(n)}$ denotes the n -th derivative of f . If f and g are each differentiable n times, then so is the product fg and

$$(fg)^{(n)} = f^{(n)}g + \binom{n}{1}f^{(n-1)}g^{(1)} + \binom{n}{2}f^{(n-2)}g^{(2)} + \cdots + g^{(n)}f.$$

101. find if the following sequences/series converge uniformly on the intervals mentioned.

(a) Sequence (f_n) where $f_n(x) = \frac{n^2 x}{1 + n^3 x^2}$, $[-1, 1]$? $[0, \infty)$?

(b) $\sum_1^\infty (x \log x)^n$ on $(0, 1)$.

(c) $\sum n^{77} e^{-nx}$ on $(0.001, \infty)$? on $(0, \infty)$?

(d) The series $\sum_1^\infty [1 - \cos(x/n)]$ converges uniformly on the interval $[-10^{10}, +10^{10}]$ but does not converge uniformly on R .

102. Similar to series which are made up of powers of x , there are important class of series which are made up of powers of n .

Let $(a_n : n \geq 1)$ be real numbers. A series of the form $\sum \frac{a_n}{n^x}$ is called Dirichlet series. We shall not complicate our life and be content with a simple instance.

Show that the series, $f(x) = \sum \frac{1}{n^x}$, converges uniformly on the interval $[1 + \epsilon, \infty)$, whatever be $\epsilon > 0$. And it does not converge uniformly on $(1, \infty)$.

Show that the series $g(x) = -\sum \frac{\log n}{n^x}$ also converges uniformly on the interval $[1 + \epsilon, \infty)$ for every $\epsilon > 0$.

Argue that f is a continuous function on $(1, \infty)$; indeed f is differentiable on $(1, \infty)$; indeed $f' = g$.

Worry is a kind of thought and memory evolved to give life direction and protect us from danger. Without its nagging whispers, we would be prone to a reckless Panglossian life style marked by drug abuse, unemployment, bankruptcy, A modest level of worry is usually best. Too much worry strands us in an agitated state of despair, anxiety and paranoia; too little leaves us without motivation and direction. Worry contributes to life's to-do list, but its relentless prompts are unpleasant and we work to diminish them by crossing items off the list.

The bottom line? Stop worrying about worry. It is good for you.

Robert Provine

103. For which values of $a \in \mathbb{R}$ is the function $f(x) = x^a \log x$ uniformly continuous on $(0, \infty)$?

104. Show

$$\frac{x}{1+x} + \frac{2x^2}{1+x^2} + \frac{4x^4}{1+x^4} + \frac{8x^8}{1+x^8} + \cdots$$

converges to $x/(1-x)$ when $|x| < 1$.

105. Show that $\frac{(1+\sqrt{5})^n - (1-\sqrt{5})^n}{2^n \sqrt{5}}$ is an integer. Here $n \in \mathbb{N}$.

106. Just like limit point of a sequence, you can define limit point of a set. Let $A \subset \mathbb{R}$. Say that a number a is a limit point of the set A if there are infinitely many points of A close to a , more precisely, given $\epsilon > 0$, there are infinitely many points of A in $(a - \epsilon, a + \epsilon)$. Verify that the following formula expresses the same thing.

$$(\forall \epsilon > 0)(\forall k) \left(\text{cardinality} \left\{ (a - \epsilon, a + \epsilon) \cap A \right\} \geq k \right).$$

What does it mean to say that a is not a limit point of the set A ?

107. (most are simple, but could be tricky, first think, then plan, then execute, then write, then read what you wrote.)

Show if x is not negative integer,

$$\sum_1^\infty \frac{1}{(n+x)(n+x+1)(n+x+2)} = \frac{1}{2(x+1)(x+2)}.$$

$$\sum_2^{\infty} \frac{\log[(1 + \frac{1}{n})^n(1+n)]}{\log n^n \log(n+1)^{n+1}} = \frac{1}{\log 4}. \quad \text{Simplify } \sum \frac{(n-1)(n+1)}{n!}.$$

$$\sum_2^{\infty} \frac{1}{n^2 - 1} = \frac{3}{4}; \quad \sum_1^{\infty} \frac{n^2 x^n}{n!} = (x^2 + x)e^x.$$

$$\sum_1^{\infty} nx^n = \frac{x}{(1-x)^2}; \quad \sum_1^{\infty} (n+1)x^n = \frac{1}{(1-x)^2}. \quad |x| < 1.$$

108. Test the following series for convergence.

$$\sum \frac{\log n}{n\sqrt{n+1}}; \quad \sum \frac{n!}{(n+2)!}; \quad \sum \frac{1}{(\log n)^{1000}}; \quad \sum \frac{1 + \sqrt{n}}{(n+1)^3 - 1}.$$

$$\sum \frac{\sin(1/n)}{n}; \quad \sum [1 - n \sin(1/n)]; \quad \sum \log(n \sin(1/n)).$$

$\sum a_n$ where a_n equals $1/n$ if n is a square and equals $1/n^2$ if n is not a square.

$\sum a_n$ where a_n equals $1/n^2$ if n is odd and $-1/n$ if n is even.

109. if $\sum a_n$ is a convergent series of positive terms, show that the series $\sum \sqrt{a_n a_{n+1}}$ converges.

110. If the series $\sum a_n$ is absolutely convergent, show that the following series are also convergent. For the second series below we assume that all the a 's are different from -1 .

$$\sum a_n^2; \quad \sum \frac{a_n}{1 + a_n}; \quad \sum \frac{a_n^2}{1 + a_n^2}.$$

111. Evaluate the following limits using L'Hopital when needed.

$$\lim_{x \rightarrow 0} \frac{\log(\cos ax)}{\log(\cos bx)}. \quad \lim_{x \rightarrow 1} \frac{\sum_1^n x^k - n}{x - 1}. \quad \lim_{x \rightarrow 0^+} \frac{x - \sin x}{(x \sin x)^{3/2}}.$$

$$\lim_{x \rightarrow a} \frac{\sqrt{x} - \sqrt{a} + \sqrt{x-a}}{\sqrt{x^2 - a^2}}. \quad \lim_{x \rightarrow 1^+} \frac{x^x - x}{1 - x + \log x} \quad \lim_{x \rightarrow \infty} x^{1/x}.$$

$$\lim_{x \rightarrow 0^+} x^{1/3}(\log x)^3. \quad \lim_{x \rightarrow 0^+} x(\log x)^2. \quad \lim_{x \rightarrow 0} (\sin x)^x.$$

$$\lim_{x \rightarrow 0^+} \frac{\log \sin x}{\log(e^x - 1)}. \quad \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right).$$

112. You know that a sequence of real numbers converges iff it is a Cauchy sequence. Similar result is true for uniform convergence of functions too. Let (f_n) be a sequence of functions on a set S . Say that the sequence is uniformly Cauchy if the following holds: given $\epsilon > 0$, there is n_0 such that $|f_n(x) - f_m(x)| < \epsilon$ for every $n, m \geq n_0$ and every $x \in S$. Show that (f_n) converges uniformly on S to some function iff it is uniformly Cauchy sequence.
113. Suppose (f_n) are differentiable functions on $(-30, +30)$ and $f_n(0) \rightarrow 17$. Suppose that $f'_n \rightarrow g$ uniformly on $(-30, +30)$. Then show that the sequence $\{f_n\}$ converges uniformly to a function f and f is continuous and f is differentiable and indeed $f' = g$.
- Hint:
 $|f_n(x) - f_m(x)| \leq |[f_n(x) - f_m(x)] - [f_n(0) - f_m(0)]| + |f_n(0) - f_m(0)|$.
 Hint: If continuous functions $\varphi_n \rightarrow \varphi$ uniformly and $x_n \rightarrow x$ then $\varphi_n(x_n) \rightarrow \varphi(x)$

$$\text{Show if } f(x) = \sum_1^{\infty} \frac{\cos nx}{n^{7/3}} \quad \text{then} \quad f'(x) = - \sum \frac{\sin nx}{n^{4/3}}.$$

Do not worry if you find some problems difficult. First do what you can. After all, as we go along we need to think of a little more challenging problems. I assure you none of them is ugly needing lots of complicated calculations. Believe me you all are capable to do most of these.

I assured you that me and TAs are ready to help you. But some of you are neither attempting the problems nor asking TAs.

I keep six honest serving men ♦ they taught me all I knew;
 Their names are What and Why and When ♦
 And How and Where and Who.
 I send them over land and see, ♦ I send them east and west;
 but after they have worked for me, ♦ I give them all a rest.

Rudyard Kipling

114. A man in a boat on one shore of a river wishes to reach a point on the other shore that is 4 miles down the stream. The river is 2 miles wide and has negligible current. He can row at 4 mph and run along the opposite bank at 8 mph. If he rows along a straight line path to a point on the opposite bank and then runs along the opposite bank to his destination, where should he land to minimize time taken to reach destination.
115. A window in the form of a rectangle surmounted by an isosceles triangle (with top side of the rectangle as base), the altitude of the triangle being $(3/8)$ -th of its base. If the perimeter of the window is 30 ft, find the dimensions of the window for admitting maximum light.
116. A wire bent in the form of a circle of radius a exerts an attractive force upon a particle on the axis of the circle (that is on the line through center and perpendicular to the plane of the circle). From the theory of attraction (let us believe it), this force is proportional to $h/(a^2 + h^2)^{3/2}$ where h is the height of the particle above the plane of the circle. At what height is the attraction maximum?
117. Taking the help of Taylor, when needed (sometimes you do not need), prove the following.
- (i) For $|x| < 1$,

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots.$$

$$\log \frac{1}{1-x} = \sum \frac{x^n}{n}. \quad \frac{1}{2} \log \frac{1+x}{1-x} = \sum_1^{\infty} \frac{x^{2n-1}}{2n-1}.$$

$$(1+x)\log(1+x) = x + \sum_2^{\infty} (-1)^n \frac{x^n}{n(n-1)}.$$

$$\begin{aligned} \log(x + \sqrt{1+x^2}) &= x - \frac{1}{2} \frac{x^3}{3} + \frac{1}{2} \frac{3}{4} \frac{x^5}{5} + \cdots \\ &+ (-1)^{n+1} \frac{1 \times 3 \times 5 \times \cdots \times (2n-1)}{2^n n!} \frac{x^{2n+1}}{2n+1} + \cdots. \end{aligned}$$

(ii) For $x \in R$,

$$(1+x)e^{-x} = 1 + \sum_2^{\infty} (-1)^{n-1} \frac{n-1}{n!} x^n$$

$$\begin{aligned} \frac{e^x - 1}{x} &= 1 + \frac{x}{2!} + \frac{x^2}{3!} + \cdots + \frac{x^{n-1}}{n!} + \cdots \\ \sin^2 x &= \frac{2x^2}{2!} - \frac{2^3 x^4}{4!} + \frac{2^5 x^6}{6!} - \cdots + (-1)^{n-1} \frac{2^{2n-1} x^{2n}}{(2n)!} + \cdots. \end{aligned}$$

$$\cos^2 x = 1 + \frac{1}{2} \sum_1^{\infty} (-1)^n \frac{(2x)^{2n}}{(2n)!}.$$

$$e^x \sin x = x + x^2 + \frac{2x^3}{3!} - \frac{4x^5}{5!} + \cdots$$

I do not think the general term is $\pm(n-1)x^n/n!$, have not checked.

118. Define $f(x)$ as follows. equals $1+x^2$ for $x \leq 0$; equals $1+x^3$ for $0 < x < 2$; equals $(1+x)^2$ for $x \geq 2$. Discuss continuity and differentiability of the function at the points 0, 1, 2.

119. Let L, M, B, C be fixed real numbers. Define a function

$$U(x) = \frac{Lx + M}{x^2 - 2Bx + C}$$

Let U_n be the n -th derivative of U . Show

$$\frac{x^2 - 2Bx + C}{(n+1)(n+2)} U_{n+2} + \frac{2(x-B)}{n+1} U_{n+1} + U_n \equiv 0.$$

Our rulers invent attractive phrases to divert your attention from the quiet destruction they are causing. 'Learning Outcome Based Curriculum' by UGC is one such and they do not even know the outcomes of courses.

Let me assure you, anyone who says outcome of Calculus 1, one of the first courses after school, is simply to learn continuous functions etc etc; is clueless and ignorant. You will learn how to formulate intuitive ideas rigorously; how to argue your case and organize proofs; how concepts appear at diverse places in diverse ways; how you scale peaks with pure thought process; and finally, prepares groundwork for future. A course should help you think independently, make you confident and even question what is being said if you disagree. Our rulers do not like this.

120. Show

$$\begin{aligned}\gamma &= \sum_1^{\infty} \left[\frac{1}{n} - \log\left(1 + \frac{1}{n}\right) \right] = 1 - \log 2 + \sum_2^{\infty} \left[\frac{1}{n} - \log\left(1 + \frac{1}{n}\right) \right] \\ &= 1 + \sum_1^{\infty} \left[\frac{1}{n+1} - \log\left(1 + \frac{1}{n}\right) \right]\end{aligned}$$

121. Show

$$\log 3 = 1 + \frac{1}{2} - \frac{2}{3} + \frac{1}{4} + \frac{1}{5} - \frac{2}{6} + \frac{1}{7} + \frac{1}{8} - \frac{2}{9} + + - + + - \dots$$

Show $\log x = \lim_{n \rightarrow \infty} n(\sqrt[n]{x} - 1)$. Apparently this is called Halley's formula (after Edmund Halley). Use $\sqrt[n]{x} = e^{\log x/n}$.

122. Verify by hand that the first four terms of $\cos(3/2)$ add to $359/(2^{10} \cdot 5)$ and all even sums are nonnegative from then on.

Show each even sum in $\sin x$ series is positive if $0 < x < 2$ and hence $\cos x$ is strictly decreasing in $[0, 2]$. Conclude $\pi > 3$. Show $e < 3$.

123. Show $[1 + (x/y)]^y < e^x$ for $x, y > 0$ and taking $x = \pi - e$ and $y = e$ deduce $\pi^e < e^\pi$.

Many problems are standard and can be found in several books. Above problems are from the amazing book by Paul Loya 'Amazing and Aesthetic Aspects of Analysis'.

124. A function f continuous on $(1, 3)$, differentiable at every point $x \neq 2$. We know that $f'(x) \rightarrow 97$ as $x \rightarrow 2, x \neq 2$. This means: given $\epsilon > 0$, there is a $\delta > 0$ such that $|f'(x) - 97| < \epsilon$ when $0 < |x - 2| < \delta$. Show f is differentiable at 2 and $f'(2) = 97$.
125. A function f defined on an interval I is said to be concave if $x, y \in I; 0 < \lambda < 1 \Rightarrow f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$. Show that f is concave iff $(-f)$ is convex.
126. Can you give a continuous function on R which is differentiable at all but one point? differentiable at all but two points? at all but 100 points? At all but integers?
There are continuous functions on R which are nowhere differentiable, that is, at each point, $f'(x)$ does not exist. If we have time we shall do later.
If $f_n \rightarrow f$ uniformly and each f_n is differentiable, do you think f must be differentiable?
127. Using product of power series show for $|x| < 1$,
- $$\frac{\log(1+x)}{1+x} = x - (1 + \frac{1}{2})x^2 + (1 + \frac{1}{2} + \frac{1}{3})x^3 - \dots$$
- $$+ (-1)^n(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n})x^n.$$
128. If $a_0 + \frac{a_1}{2} + \frac{a_2}{3} + \dots + \frac{a_n}{n+1} = 0$ show that the polynomial $p(x) = \sum_0^n a_k x^k$ has at least one root between in $(0, 1)$
129. Generalizing our discussion of $x^m \sin(1/x^n)$ function; fix $a \in R$ and $c > 0$. Define f on $[-1, 1]$ by $f(0) = 0$ and $f(x) = x^a \sin(\frac{1}{|x|^c})$ for $x \neq 0$.
Show f is continuous iff $a > 0$.
Show $f'(0)$ exists iff $a > 1$.
Show f' is bounded iff $a \geq c + 1$.
Show $f''(0)$ exists iff $a > c + 2$.
Show f'' is bounded iff $a \geq 2c + 2$.
Show f'' is continuous iff $a > 2c + 2$.
130. Calculate $\lim_{x \rightarrow 0} (\frac{1}{x} - \frac{1}{\sin x})$.

The most important scientific tool of all is not anything you can buy.
It is your own mind. Your thoughts and ideas are the keys that can unlock
the mysteries. In the search for understanding; "questions" are perhaps
the most powerful "force" of all. – Jack R White

May all feel Happiness ◇ May all have sound Health.

May all see the Auspicious ◇ May no one feel sorrow.

Upanishads

131. A sequence of continuous functions (f_n) converges uniformly on $[a, b]$ to f . Show $\int_a^b f$ is defined and $\int_a^b f_n \rightarrow \int_a^b f$.
Consider the unit interval $[0, 1]$. For $n > 1$ consider the function f_n : zero at zero, increases linearly to 1 when $x = 1/n$ and falls linearly to zero when $x = 2/n$ and remains zero from $2/n$ to 1. Show $f_n(x) \rightarrow 0$ for each x . But $\int_0^1 f_n$ is not converging to zero. Does this contradict above?
132. Define function f on $[0, 1]$ by $f(x) = 1$ except when $x \in \{1/n : n \geq 1\}$ and $f(1/n) = 2$ for all $n \geq 1$. Show f is Riemann integrable.
133. Show (Do not differentiate right side, try to go from left to right).

$$\int \frac{1}{x} \log x = \frac{1}{2}(\log x)^2.$$

$$\int e^{ax} \sin bx = \frac{1}{a^2 + b^2} e^{ax} [a \sin bx - b \cos bx]$$

$$\int e^{ax} \cos bx = \frac{1}{a^2 + b^2} e^{ax} [a \cos bx + b \sin bx]$$

134.

$$u_n = \int_0^{\pi/4} \tan^n x dx. \quad \text{show} \quad u_{n+2} + u_n = \frac{1}{n+1}.$$

135. Show (here $m, n \geq 1$ integers)

$$\int_{-\pi}^{+\pi} \sin mx \cos nx = 0 = \int_{-\pi}^{+\pi} \sin nx = \int_{-\pi}^{+\pi} \cos nx.$$

$$m \neq n \Rightarrow \int_{-\pi}^{+\pi} \sin mx \sin nx = 0 \quad \int_{-\pi}^{+\pi} \cos mx \cos nx = 0$$

$$\int_{-\pi}^{+\pi} \sin nx \sin nx = \pi \quad \int_{-\pi}^{+\pi} \cos nx \cos nx = \pi$$

Define $\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} fg$ for continuous functions f, g on $[-\pi, \pi]$. We can restate above equations as: If $f, g \in \{1; \sin nx, \cos nx, n \geq 1\}$, then $\langle f, g \rangle$ equals one or zero according as f, g are same or different. Do you see flavour of orthonormal vectors?

136. Show

$$(-1)^n \int_{-1}^1 (x^2 - 1)^n = \frac{2^{2n+1}(n!)^2}{(2n+1)!}.$$

Let $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$. These are called Legendre Polynomials. Show

$$\begin{aligned} \int_{-1}^1 P_n(x) P_m(x) dx &= 0; & m \neq n. & \quad \int_{-1}^1 P_n^2(x) dx = \frac{2}{2n+1}. \\ \int_{-1}^1 x^m P_n(x) dx &= 0; & m < n; & \quad \int_{-1}^1 x^n P_n(x) dx = ?. \end{aligned}$$

Define $\langle f, g \rangle = \int_{-1}^1 fg$ for continuous functions f, g on $[-1, 1]$. Then there are constants c_n such that $\langle c_n P_n, c_m P_m \rangle$ is one or zero according as $m = n$ or not. Orthonormal vectors again?

137. Show: $D^n(e^{-x^2/2}) = (-1)^n u_n(x) e^{-x^2/2}$; u_n is a polynomial of degree n .

Show $u_{n+1} = xu_n - u'_n$.

Using $D(e^{-x^2/2}) = -xe^{-x^2/2}$ and Leibnitz, show $u_{n+1} = xu_n - nu_{n-1}$

Show: $u_0 \equiv 1$; $u_1 \equiv x$; $u_2 = x^2 - 1$; $u_3 = x^3 - 3x$

$u_4 = x^4 - 6x^2 + 3$; $u_5 = x^5 - 10x^3 + 15x$.

These (or $c_n u_n$ for some constants c_n depending on your purpose) are called Hermite Polynomials. These are also orthonormal vectors, but you should wait for integration over unbounded intervals.

138.

$$\binom{n}{k} = \frac{1}{(n+1) \int_0^1 x^k (1-x)^{n-k} dx}.$$

Calculate

$$\int \frac{x}{(x^2 + 1)^n}; \quad n = 1, 2, 3, \dots$$

139. f is a continuous function on $[a, b]$ and $\int_a^x f = 0$ for $a < x < b$. Show that $f \equiv 0$.

f is a continuous function on $[0, \infty)$ and $f(x) = \int_0^x f(t) dt$ for all $x \geq 0$. Show that $f \equiv 0$. (Calculate derivative and see).

140. Let f be a non-negative continuous function on $[a, b]$ with maximum value M . Show

$$\left[\int_a^b f^n(x) dx \right]^{1/n} \rightarrow M.$$

(see if $\leq M$ and $\geq (M - \epsilon)$, but does the limit exist?)

141. Show

$$\lim_{k=1}^n \frac{n}{k^2 + n^2} = \frac{\pi}{4}. \quad \lim_{k=1}^n \frac{1}{\sqrt{n^2 + k^2}} = \log(1 + \sqrt{2}).$$

142. Calculate

$$\int \frac{\sqrt{1 + \sqrt{y}}}{\sqrt{y}} dy; \quad \int_0^9 \sqrt{1 + \sqrt{y}} dy.$$

143. Let f be a continuous function on R such that $\lim_{x \rightarrow \pm\infty} f(x) = 0$. Show that f is uniformly continuous on R . Show that the same holds if $\lim_{x \rightarrow \pm\infty} f(x) = 59$.

144. Find derivatives:

(Sometimes taking logarithms and differentiating helps). Take an interval in which the given function is defined and do.

$$f(x) = \frac{ax^2}{\sqrt[3]{x}} + \frac{b}{x\sqrt{x}} - \frac{\sqrt[3]{x}}{\sqrt{x}} \quad f(x) = \frac{2x^4}{b^2 - x^2} \quad f(x) = (x \cot x)^2$$

$$f(x) = \sqrt{\frac{1+x}{1-x}} \quad f(x) = \sqrt{x + \sqrt{x + \sqrt{x}}} \quad f(x) = x^{\log x}$$

$$f(x) = \tan \frac{1 - e^x}{1 + e^x} \quad f(x) = \sin \sqrt{1 - 2^x} \quad f(x) = \sqrt[3]{\frac{x(x^2 + 1)}{(x - 1)^2}}$$

$$f(x) = \frac{\sqrt[5]{(x - 1)^2}}{\sqrt[4]{(x - 2)^3} \sqrt[3]{(x - 3)^7}}$$

Generally our exercises are not ugly. But some of the above derivatives do need calculations and may be not nice. As long as you are understanding the material and are not lazy, you should be able to do easily.

It matters little who first arrived at an idea, rather what is significant is how far that idea can go.

Sophie Germain

Doing Math is like being lost in a jungle and trying to use all the knowledge you can gather to come up with new tricks, and with some luck you may find a way out.

M Mirzakhani

With every act of thinking, something permanent, substantial enters our soul.

Riemann

The real satisfaction from mathematics is in learning from others and sharing with others. All of us have clear understanding of a few things and murky concepts of many more.

Bill Thurston

145. Show $\Gamma(n) = (n-1)!$ for $n \geq 1$. (Induction)
 Show $\beta(m, n) = \frac{(m-1)!(n-1)!}{(m+n-1)!} = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ (m, n are integers. ≥ 1).
146. (i) Show $\sin : [-\pi/2, +\pi/2] \rightarrow [-1, +1]$ is a continuous bijection. Its inverse is called 'arcsine' or $\sin^{-1} : [-1, +1] \rightarrow [-\pi/2, +\pi/2]$. Calculate its derivative. Calculate $\int_0^1 \sqrt{1-u^2} du$.
- (ii) Show $\cos : [0, \pi] \rightarrow [-1, 1]$ is a continuous bijection. Its inverse is called 'arccosine' or $\cos^{-1} : [-1, +1] \rightarrow [0, +\pi]$. Calculate its derivative. Show $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$.
- (iii) Show $\tan : (-\pi/2, +\pi/2) \rightarrow (-\infty, +\infty)$ is a continuous bijection. Its inverse is called 'arctan' or $\tan^{-1} : \mathbb{R} \rightarrow (-\pi/2, +\pi/2)$. Calculate its derivative. Calculate $\int_{-\infty}^{+\infty} \frac{1}{1+x^2} dx$.
- (iv) Show $\sinh : \mathbb{R} \rightarrow \mathbb{R}$ is continuous bijection. Show $\sinh^{-1}(x) = \log(x + \sqrt{x^2 + 1})$.
- (v) Show $\cosh : [0, \infty) \rightarrow [1, \infty)$ is continuous bijection. Show $\cosh^{-1}(x) = \log(x + \sqrt{x^2 - 1})$.
147. Is the function $f(x) = \sin(1/x)$ uniformly continuous on the interval $(0, 1)$? on $(1, \infty)$? Do the same for $g(x) = x \sin(1/x)$?
148. Integral without limits means a primitive. If

$$I_n = \int \frac{1}{(x^2 + 1)^n} dx$$

reduce I_n to I_{n-1} by writing $I_n = \int \cos^{2n-3} x (\sin x)'$ to deduce

$$I_n = \frac{x}{(2n-2)(1+x^2)^{n-1}} + \frac{2n-3}{2n-2} I_{n-1}$$

149. Which of the functions are integrable on $(0, \infty)$; on $(1, \infty)$; on $(0, 1)$.

$$\frac{x}{\sqrt{x}(1+x)}; \quad \frac{x}{1+x^2}; \quad \frac{\sqrt{x+1}-\sqrt{x}}{1+x}; \quad \frac{1}{x+x^4}.$$

150. Evaluate $\int_0^1 \sqrt{1+\sqrt{x}} dx$. Put $1+\sqrt{x} = y$ so that $dx = 2(y-1)dy$.

151. Let v and u be differentiable functions on an interval and are strictly positive and $v > 1$. Let $f(x) = \log_{v(x)} u(x)$, that is $\log u(x)$ to the base $v(x)$. Calculate its derivative.

152. Discuss convergence of the following integrals.

$$\begin{aligned} & \int_1^\infty \sin(1/x) dx; \quad \int_1^\infty \sin(1/\sqrt{x}) dx; \quad \int_1^\infty \sin(1/x^2) dx. \\ & \int_0^1 \sin(1/x) dx; \quad \int_0^1 \sin(1/\sqrt{x}) dx; \quad \int_0^1 \sin(1/x^2) dx. \\ & \int_0^\infty e^{-x} \frac{\sin x}{x} dx; \quad \int_1^\infty e^{-2x} \cos(4x) dx; \quad \int_0^1 \frac{\cos 4x - \cos 5x}{x} dx. \\ & \int_0^1 x^{n-1} \log x dx; \quad \int_0^\infty e^{-x^2} dx; \quad \int_{-\infty}^\infty e^{(x-5)^2/33} dx. \end{aligned}$$

153. (Do not open pen) $\int_0^{\pi/2} \log \sin x dx = -\frac{\pi}{2} \log 2 = \int_0^{\pi/2} \log \cos x dx$
 $\int_0^\pi \log(1 - \cos x) dx = -\pi \log 2 = \int_0^\pi \log(1 + \cos x) dx.$

154.

$$\text{Evaluate:} \quad \int_0^1 \frac{x^{2n}}{\sqrt{1-x^2}} \quad \int_0^1 \frac{x^{2n+1}}{\sqrt{1-x^2}}$$

155. This and next Exercise OPTIONAL. Fine if you do NOT attempt.

$$\begin{aligned} & \int \frac{1}{x(x+1)(x+2)(x+3) \cdots (x+n)} = \\ & \frac{1}{n!} \left[\log x - \binom{n}{1} \log(x+1) + \binom{n}{2} \log(x+2) - + \cdots \pm \binom{n}{n} \log(x+n) \right] \end{aligned}$$

156. $\int_\alpha^\infty \frac{f(x)}{x} dx$ converges for every $\alpha > 0$ and $f(x) \rightarrow L$ as $x \rightarrow 0$. Let $0 < a < b$. show $\int_0^\infty \frac{f(ax)-f(bx)}{x}$ converges and has value $L \log(b/a)$.

$$\text{Hint: } \int_\alpha^\infty \frac{f(ax)-f(bx)}{x} = \int_{a\alpha}^{b\alpha} \frac{f(x)}{x} = L \log(b/a) + \int_{a\alpha}^{b\alpha} \frac{f(x)-L}{x}$$

Show last term goes to zero as $\alpha \rightarrow 0$.

Deduce

$$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} = \log(b/a) = \int_0^\infty \frac{\cos(ax) - \cos(bx)}{x}$$

Having exhausted all ideas and having achieved success in creating hatred against some religious faiths, our rulers have turned their attention now to the different states. This was flagged off by the PM, inciting the people of a state against us with a deceptive, offensive and irresponsible narrative. This was taken forward in Delhi by assaulting people from south. You should be careful outside class room because we have rulers who dislike truth and science. In the class room you should maintain scientific spirit and question if you disagree with any mathematics being done.

157. Trace the curve showing where it crosses x -axis, increasing, decreasing, maxima and minima:

$$f(x) = x^2 - x - 12; \quad f(x) = x^3 - 2x^2 + x$$

$$f(x) = x^3 + 3x^2 - x - 3; \quad f(x) = x^{100} - 1$$

$$f(x) = x \sin(1/x); \quad f(x) = \frac{1}{x} \sin x$$

158. First sketch the two curves below and then find the area enclosed between the two curves $y = x^3 - 12x$; $y = x^2$.
Same as above for the following two curves: $y = x^3 - x$ and the tangent to the curve $y = x^3 - x$ at $x = -1$.
Same as above for the following curves:

$$x + y = 1; \quad x + y = -1; \quad x - y = 1; \quad x - y = -1.$$

(such problems involve work and time. You should take interest and practice. They will not be for exam and you should not be tense,)

159. All the functions f_n and f below are assumed to be continuous on $[0, \infty)$. suppose that $f_n \rightarrow f$ uniformly.
If each f_n is integrable, then is it necessary that f is integrable?
if each f_n is integrable and f is also integrable then is it necessary that $\int_0^\infty f_n \rightarrow \int_0^\infty f$?
You have done similar problem earlier for a bounded closed interval.

160. show that the series

$$\sum_{-\infty}^{\infty} e^{-n^2}; \quad \sum e^{-n^2+1000n+10000000}$$

$$\sum \frac{(\log n)^{89}}{n^{1.01}}; \quad \sum \left(\sin \frac{1}{n} \right)^{1.001}; \quad \sum \left[\binom{2n}{n} 2^{-2n} \right]^{2.01}$$

161. Show that

$$0 < x \leq 1 \Rightarrow |x \log x| \leq 1/e.$$

Let $a > 0$. Prove that the series

$$1 + a(x \log x) + \frac{a^2}{2!}(x \log x)^2 + \frac{a^3}{3!}(x \log x)^3 + \dots$$

converges to x^{ax} uniformly on $(0, 1]$. Deduce

$$\int_0^1 x^{ax} dx = \sum_0^\infty (-1)^n \frac{a^n}{(n+1)^{n+1}}.$$

162. Show that for $|x| < 1$,

$$\int_0^x \log \frac{1+t}{1-t} dt = \sum_1^\infty \frac{x^{2n}}{n(2n-1)}.$$

163. Here are special cases of the Binomial theorem. Verify

$$\frac{1}{\sqrt{1+x}} = 1 - \frac{x}{2} + \frac{1 \times 3}{2!} \left(\frac{x}{2} \right)^2 + \frac{1 \times 3 \times 5}{3!} \left(\frac{x}{2} \right)^3 + \dots$$

$$\frac{1}{\sqrt{1-x^2}} = 1 + \frac{x^2}{2} + \frac{1 \times 3}{2!} \frac{x^4}{2^2} + \frac{1 \times 3 \times 5}{3!} \frac{x^6}{2^3} + \frac{1 \times 3 \times 5 \times 7}{4!} \frac{x^8}{2^4} + \dots$$

Write down special cases of Binomial theorem: $1/(1+x)^2$ and $1/(1+x)^3$
NOT using $\binom{m}{n}$ with negative m .

164. Fix $y \in R$ and $t > 0$. Calculate

$$\int_{-\infty}^{\infty} e^{-(x-y)^2/2t}$$

Let $p(t, x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}$ defined for $t > 0$ and $x \in R$.

For each fixed $t > 0$, find second derivative of p as a function of x .

For each fixed $x \in R$, find first derivative of p as a function of t .